# Spec: multi-branch decoder that learns the cut

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Replace the fixed (a, b) decoder with a minimum over K learned smooth branches, so that training decides where travel-time ridges are and how many competing route families exist. This document is written for an implementer who already knows the smooth-graph method, graph_gram and the G-Shape-NTFields code.

## 1. Summary

The fixed decoder D(a, b) can only place a ridge at the rotational half-turn. With obstacles, ridges move, vanish or appear (for example going over versus under a bar), and the paper's stage E shows graph_gram falling behind for this reason. This spec replaces the fixed decoder with a minimum over K smooth branches. The ridge (the cut) is then wherever two branches tie, so its location and the number of competing route families are learned.

**Keep from the existing pipeline:**

- the G-Shape encoder with Fourier features, Lipschitz-normalised layers and the input skip [e; G(x)];
- the intrinsic endpoint derivatives $c_x$, c_y (Proposition 1 of the implementation report) and the endpoint Eikonal loss;
- the EMA-target finite-step Bellman loss and its candidate control set;
- regular-pair masking defined from input geometry, never from predicted quantities;
- the planners, evaluation harness and stage-E environments.

**New:** K branch heads, a min decoder with an annealed soft-min, band-limited per-branch losses, a distinctness loss at ties, and branch management (usage tracking, revival, pruning, merging).

**Non-goals for version 1:** no change to the encoder; no homotopy labels or H-signatures; no claim of novelty until a literature check (min-plus networks by Darbon, Dower and Meng are the closest known prior work).

## 2. Background and notation

Notation follows the implementation report: poses $x = (t_x, R_x)$, embedding $G(x) = (t_x, \mathrm{vec}\, R_x) \in \mathbb{R}^{12}$, ellipsoidal body-twist limit with scales $s_t$ and $s_r$, scaled body frame $E_i = S\, e_i$, endpoint gradients $c_x$ and $c_y$ in that frame, $\tau_{ref} = L_{ref} / s_t$. The following facts drive the design.

1. **Kink structure.** The true travel time T has a cone at the goal and otherwise only concave kinks (ridges, local shape Λ). It never has a V-shaped kink away from the goal. This is the viscosity-solution condition.
2. **Local structure.** Away from the goal and from conjugate points, T is locally the minimum of finitely many smooth functions (Cannarsa and Sinestrari, semiconcave functions). Each smooth piece is the arrival time along one family of locally optimal routes.
3. **Two kinds of ridges.** Topological ridges come from different homotopy classes tying (the ±q half-turn on SO(3), left versus right of a pole). Geometric ridges come from two routes in the same class tying (over versus under a bar). On SE(3) most obstacle ridges are geometric, so homotopy labels alone do not explain them.
4. **Sheet.** A sheet is the arrival-time function of one route family. It is smooth on its own domain, satisfies the Eikonal equation there, and is valid beyond the region where it is the minimum (the long way around is still a route).
5. **Monodromy.** A sheet cannot in general be a single continuous function on the whole configuration space: following it around a loop (for example a full rotation) returns a different value. A branch in this design is therefore a smooth function on the base space that equals a sheet only on its competitive band and is merely $\geq T$ elsewhere. Its own discontinuity of identity is hidden where it is not competitive.

## 3. Formulation

The decoded time is the minimum of K branches, each the square root of a smooth nonnegative function:

```
T_\theta(x,y)=\min_{k=1,\dots,K} T_k(x,y),\qquad T_k=\sqrt{Q_k},\quad Q_k\ge
0 \text{ smooth}
```

This partitions pair space into regions and a cut:

```
R_k=\{T_k=T_\theta\},\qquad \mathcal C=\{\,T_i=T_j=T_\theta,\ i\neq
j\,\},\qquad B_k=\{\,T_k\le T_\theta+\Delta\,\}
```

$R_k$ is where branch k wins, 🚫 is the learned ridge set, and $B_k$ is branch k's competitive band with margin Δ. Learning the cut means learning this partition together with a smooth field on each piece.

**Each branch is meant to be a sheet on its band.** Inside $B_k$, $T_k$ must be the true arrival time of one route family: it satisfies the Eikonal equation and decreases at unit rate along its own route. Outside $B_k$ it only has to stay at or above Tθ, which the minimum already guarantees.

**The losing branch places the cut.** The tie $T_i = T_j$ happens where branch j catches up with branch i. If only the active branch is trained, the losing one drifts and the cut lands in the wrong place. This is why every loss in section 5 that shapes a branch is applied on its band, not only on its winning region.

**Worked example (circle, goal at 0°, unit speed).** Without obstacles, two branches suffice: one equals |δ| near the goal and continues as the counterclockwise time δ slightly past 180°, the other equals the clockwise time 360° − δ near 180°; both are smooth on the circle and rise above T where they are not competitive. They tie at 180°, which is the ridge. If the arc [90°, 120°] is blocked and both endpoints lie in the free arc, only one route exists, one branch suffices, and no ridge appears. The learned branch count should change accordingly.

## 4. Architecture

One shared siamese encoder feeds K cheap heads; the decoder is a minimum over the heads.

```
x, y  -> G(x), G(y) -> shared G-Shape encoder -> e_x, e_y   (with input skip
[e; G])
      -> head 0 (direct):  Q_0(e_x, e_y)
      -> head k (detour):  Q_k(e_x, e_y),  k = 1..K-1
      -> T_k = sqrt(Q_k) -> soft-min at temperature tau (training) / hard min
(evaluation) -> T_theta
```

**Direct head (k = 0) carries the goal cone.** It must vanish quadratically at x = y and only there:

$$Q_0=\Big(\lVert A_0(e_x-e_y)\rVert^2+\rho\,\lVert G(x)-G(y)\rVert^2\Big)\exp\!\big(g_0(x,y)\big),\qquad g_0=2\tanh(\cdot)$$

The ρ term (default 0.01) guarantees $Q_0 > 0$ whenever $x \neq y$; the bounded factor $\exp(g_0)$ keeps the head flexible away from the goal.

**Detour heads (k ≥ 1) have a positive value at x = y** (a route that leaves and returns), so they never compete at the goal:

$$Q_k=\operatorname{softplus}\big(f_k(e_x,e_y)\big)^2,\qquad k\ge 1$$

**Symmetry.** Build every head from exchange-symmetric features, for example $s = e_x + e_y$ and $d = (e_x - e_y)\odot(e_x - e_y)$, or average the raw head over swapping x and y. This is valid because reversing a route keeps it in the same family.

**Head forms (a config switch, to be ablated):**

| Form | Branch definition | Use |
|---|---|---|
| free | MLP heads as above | default for stages 0 to 2 |
| folded | per-branch graph_gram latents ($a_k$, $b_k$) with $Q_k = a_k/s_t^2 + \varphi(b_k)/s_r^2$; detours add $\text{softplus}(c_k)^2$ to $a_k$ | stage 3; keeps the exact half-turn ridge inside each branch |
| folded-metric | as folded, but $a_k$ from the metric-head distance as in graph_metric | stage 3 default, since stage E showed Euclidean a is too restrictive with obstacles |

**Temperature.** During training, combine branches with a soft-min; at evaluation and planning, use the hard min.

$$T_\tau=-\tau\log\sum_{k}e^{-T_k/\tau},\qquad w_k=\operatorname{softmax}_k(-T_k/\tau)$$

Default K = 4 for stages 0 to 2 and K = 6 for stage 3. K is an upper bound; the number of surviving branches is a result, not a setting.

# 5. Losses

The total objective is a weighted sum of the terms below; $c_x^{(k)}$ and $c_y^{(k)}$ are the endpoint gradients of branch k, computed exactly as $c_x$ and c_y are today. All derivative losses use regular pairs only (mask from input geometry). Let $T_{(1)} \leqslant T_{(2)}$ be the two smallest branch values and $g = T_{(2)} - T_{(1)}$ the gap.

**Decoded Eikonal** (hard min, away from the cut):

```
L_{E}^{\mathrm{dec}}=\mathbb E\Big[\mathbf 1\{g>3\tau\}\big((\lVert
c_x\rVert^2-1)^2+(\lVert c_y\rVert^2-1)^2\big)\Big]
```

**Sheet Eikonal** (every branch on its band; this is what places the cut correctly):

```
L_{E}^{\mathrm{sheet}}=\frac{\mathbb E\Big[\sum_k \mathbf 1\{(x,y)\in
B_k\}\big((\lVert c_x^{(k)}\rVert^2-1)^2+(\lVert c_y^{(k)}\rVert^2-
1)^2\big)\Big]}{\mathbb E\big[\sum_k \mathbf 1\{(x,y)\in B_k\}\big]}
```

**Bellman on the decoded time:** the existing EMA-target finite-step loss, applied to Tθ with the hard min in the target. It selects the global solution and removes branches that claim shortcuts no admissible motion realises.

**Sheet consistency** (each branch drops at unit rate along its own route): for $(x, y) \in B_k$, step along the branch's own steepest descent and compare with the EMA copy of that branch:

```
u_k=-\frac{c_x^{(k)}}{\lVert c_x^{(k)}\rVert},\qquad
x'=x\exp\!\big(h_s\,\widehat{Su_k}\big),\qquad L_{\mathrm{cons}}=\mathbb
E\Big[\big(T_k(x,y)-\operatorname{sg}[\,h_s+T_k^{-}
(x',y)\,]\big)^2\Big]/\tau_{\mathrm{ref}}^2
```

Apply it only when x′ is collision-free and (x′, y) stays in the band.

**Distinct at ties** (a real ridge has different gradients on its two sides; equal values and equal gradients mean duplicate branches):

```
L_{\mathrm{dist}}=\mathbb E\Big[\sum_{i<j}\mathbf 1\{\lvert T_i-
T_j\rvert<\eta,\ (x,y)\in B_i\cap
B_j\}\,\operatorname{relu}\big(\cos(c_x^{(i)},c_x^{(j)})-\gamma\big)^2\Big]
```

Average the same term over c_y.

**Optional terms:**

- Lower bound (stage 3): $\max(0, T_{free} - T\theta)^2 / \tau_{ref}^2$.
- Route anchors (stages 0 to 2, ablation only): given the true times $\{\tau_r\}$ of the distinct routes at a pair, $L_{route} = \Sigma_r \min_k (T_k - \tau_r)^2 / \tau_{ref}^2$. It is permutation-invariant, so no branch labels are needed.

| Term | Applies to | Default weight |
|---|---|---|
| Decoded Eikonal | $T\theta$, pairs with gap $g > 3\tau$ | 1.0 |
| Sheet Eikonal | each branch on its band $B_k$ | 1.0 |
| Bellman | $T\theta$ | 1.0 |
| Sheet consistency | each branch on $B_k$, step $h_s = 0.05\,\tau_{ref}$ | 0.5 |
| Distinct at ties | pairs of branches within $\eta = 0.02\,\tau_{ref}$, $\gamma = 0.95$ | 0.1 |
| Lower bound | $T\theta$, stage 3 only | 1.0 |
| Route anchors | branches, ablation only | 0 (1.0 in the ablation) |

The goal condition needs no loss: the direct head is exactly zero at $x = y$ and every detour head is positive there.

## 6. Training procedure

The training loop is the existing one with a temperature schedule, a branch-management step and one extra sampling stream.

1. **Initialise branches differently.** Use different random seeds per head, and give each detour head a different initial bias in $f_k$ so the branches start at different heights. Identical initialisation never separates.
2. **Anneal the temperature.** Start at $\tau_0 = 0.1\,\tau_{ref}$, decay log-linearly to $\tau_{end} = 10^{-3}\,\tau_{ref}$ over the first 60 % of training, then hold. Early on every nearly-best branch receives gradient; at the end the cut is sharp.
3. **Set the band.** $\Delta = 0.1\,\tau_{ref}$ by default. The band must reach past the cut on the losing side, so do not make it smaller than the expected rounding width of the learned ridges.
4. **Sample near cuts.** Add a stream (default 20 % of each batch) of pairs where the current gap $g$ is small, found by evaluating a candidate pool and keeping the smallest gaps. Without it, cut regions are rare and train slowly. In stage 3 keep the existing near-half-turn stream as well.

5. **Track usage.** Every 500 steps, compute $u_k$ = the fraction of a fixed held-out pool where branch k is the hard minimum.
6. **Revive dead branches.** If $u_k < 1\,\%$ for 3 consecutive checks, copy the head of the branch with the largest sheet-Eikonal residual in its band, add weight noise (relative scale 0.1), raise its output by a constant (0.2 $\tau$_ref), and reset its optimiser state. This mirrors reviving dead centroids in k-means.
7. **Prune.** A branch revived twice that is still dead is masked out permanently. Log every revival and prune.
8. **Merge duplicates.** If two branches satisfy the duplicate condition of section 5 (values within $\eta$ and gradient cosine above $\gamma$) on more than 50 % of the pairs where both are in band, mask the one with lower usage.
9. **Report the surviving count** at the end of training; it is the learned number of competing route families in the training distribution.

**Optional curriculum for stage 3:** train obstacle-free first, then add obstacles and unfreeze extra detour heads. **Optional warm start:** for a few thousand steps use the route-anchor loss, then switch it off.

## 7. Failure modes and diagnostics

Each failure has a symptom you can log during training; check these before tuning anything else.

| Failure | Symptom | Remedy |
|---|---|---|
| Winner takes all | one branch has usage near 100 % from early on | soft-min schedule, band losses, revival |
| Dead branch | usage below 1 % for several checks | revival, then pruning (section 6) |
| Collapse | branches nearly equal in value and gradient over large regions | different initialisation, distinctness loss, merging |
| Misplaced cut | learned cut offset from the true ridge; losing branch has a large sheet-Eikonal residual near the cut | widen $\Delta$, raise the sheet-Eikonal weight, more near-cut samples |

| Failure | Symptom | Remedy |
|---|---|---|
| Fake cut | ties where the two gradients are nearly parallel | distinctness loss; merge the pair |
| False shortcut | a branch below the true time in a region where its route does not exist | Bellman weight; check that Bellman candidates are collision-checked |
| Continuous ties | behind a round obstacle a whole circle of routes ties along a line; finitely many branches only approximate it | expected; the error is confined to a thin region, report it rather than fix it |
| Saturated temperature | Eikonal residual stays high at the cut late in training | check that $\tau$ actually reached $\tau_end$; the decoded Eikonal must be masked by the gap |

**Log every 500 steps:** usage $u_k$ per branch; mean sheet-Eikonal residual per branch on its band; fraction of pairs with gap below $3\tau$; histogram of gradient cosines at near-ties; revival and prune events; current $\tau$.

**Plot at checkpoints (stages 0 to 2):** on fixed 2D slices, colour each point by its active branch, overlay the learned cut and the ground-truth ridges, and plot each branch's value along one line that crosses a ridge.

## 8. Experiments

Run the stages in order; each has a ground truth that the next one lacks. Do not start a stage until the previous one passes its gates (section 9).

| Stage | Setting | Ground truth | Expected branch count |
|---|---|---|---|
| 0a | circle, unit speed, no obstacle | closed form, $T = \min(\delta, 360° - \delta)$ | 2 |
| 0b | circle with arc [90°, 120°] blocked, both endpoints in the free arc | closed form, distance along the free arc | 1 |

| Stage | Setting | Ground truth | Expected branch count |
|---|---|---|---|
| 1a | 2D point robot in $[-1, 1]^2$, disk of radius 0.3 at the centre | fast marching | 2 near the disk's shadow |
| 1b | same, horizontal bar from (−0.5, 0) to (0.5, 0), thickness 0.05 | fast marching | 2 |
| 1c | same, two disks | fast marching | 3 or more |
| 2 | SE(2) rectangle 0.3 × 0.1 with the bar of 1b | fast marching on (x, y, θ) | rotation ridge plus obstacle ridges |
| 3a | obstacle-free SE(3), free heads, K = 2 | analytic T* | 2 (the ±q sheets) |
| 3b | SE(3) stage-E environments | none; planning metrics | reported |

**Stage 0.** Use (cos δ, sin δ) inputs and a small encoder. This checks the mechanics: the cut appears at 180° in 0a and disappears in 0b.

**Stage 1 ground truth.** Solve the Eikonal equation by fast marching (for example scikit-fmm) on a 512 × 512 grid for each of 32 fixed evaluation goals, with obstacle cells masked. To get the individual sheets, run fast marching again with a thin barrier from the obstacle to the domain boundary on one side; this forces routes around the other side and gives that side's arrival time. T is the minimum of the barrier solutions, the true ridge is where they tie with different gradients, and the larger value is the second-best route time. In 2D these obstacles create real homotopy classes, so the barrier trick is exact.

**Stage 2 ground truth.** The ellipsoidal body-twist limit gives the metric $\|\dot{t}\|^2/s_t^2 + \dot{\theta}^2/s_r^2$, which is Euclidean in the rescaled coordinates (x/s_t, y/s_t, θ/s_r). Isotropic fast marching therefore works on a 128 × 128 × 64 grid with θ periodic; confirm that the chosen solver supports periodic boundaries. Build the occupancy grid by collision-checking the rectangle at each (x, y, θ). Find ridges as cells where neighbouring gradients turn by more than 60°.

**Stage 3a** asks whether free heads can discover the topological half-turn ridge without the fold. **Stage 3b** uses the paper's T-, L- and A-shape environments with the unmodified Shape-NTFields planner; also rerun the obstacle-free benchmark (stages A to D) to check for regressions.

# 9. Metrics, baselines, ablations and gates

The method's claim lives near ridges, so the headline metric is error inside a band around the true ridges, not overall error.

**Metrics (stages 0 to 2, against ground truth):**

- **Ridge-band error:** median and 95th-percentile relative time error over points within distance r of a true ridge (r = 0.05 in scaled units); also report overall error.
- **Cut location:** mean distance from learned cut points to the true ridge set, and from true ridge points to the learned cut (both directions).
- **Branch count:** surviving branches versus the true number of route families.
- **Second-best accuracy (stages 0 to 1):** error of the second-smallest branch against the second-best route time from barrier fast marching.
- **Eikonal residual near ridges** of the decoded field.

**Metrics (stage 3):** planning success, cost ratio and collisions as in the paper, plus **branch switches per rollout** when the planner commits to a branch and switches only if another is lower by a margin (0.05 $\tau$_ref).

**Baselines** (same encoder, data, steps and seeds): K = 1 single-head field; graph_gram; graph_metric; the metric head.

**Ablations:** sheet losses applied only to the active branch (no band); no distinctness loss; no revival or pruning; hard min from the start; free versus folded versus folded-metric heads; with and without route-anchor warm start.

**Provisional gates** (3 seeds each):

| Stage | Gate to pass |
|---|---|
| 0 | branch count exactly 2 (0a) and 1 (0b); cut within 1° of 180° in 0a; median relative error below 0.5 % |
| 1 | ridge-band median error at most half that of the K = 1 baseline in at least 2 of 3 environments; correct branch count in at least 2 of 3 |
| 2 | ridge-band error better than K = 1; obstacle ridges and the rotation ridge both recovered on inspection |

| Stage | Gate to pass |
|---|---|
| 3a | a cut forms at relative angle π with two surviving branches |
| 3b | report against graph_metric (81.2 % mean success) and the metric head (81.0 %); no gate, honest comparison |

If a gate fails, report it with the diagnostics of section 7 rather than tuning silently.

## 10. Deliverables, defaults and what not to do

**Deliverables:**

- A branched head module next to the existing graph heads (suggested: models/graph_method/branched/), selectable by config, with the three head forms of section 4.
- Configs per stage, with the defaults below.
- Ground-truth generators for stages 0 to 2, including the barrier fast-marching sheets.
- Unit tests: T is unchanged when branches are permuted; $T(x, x) = 0$ exactly and every detour head is positive at $x = y$; $T(x, y) = T(y, x)$ to $10^{-6}$; the soft-min approaches the hard min as $\tau \to 0$; per-branch $c_x$ matches central finite differences; band and gap masks use detached values; revival resets optimiser state.
- A results report: tables for section 9, branch-region maps with cut overlays for stages 0 to 2, usage curves, and the revival and prune log.

**Defaults:**

| Setting | Default |
|---|---|
| K (stages 0–2 / 3) | 4 / 6 |
| ρ in the direct head | 0.01 |
| Temperature τ | $0.1\,\tau_ref \to 10^{-3}\,\tau_ref$, log-linear over the first 60 % |
| Band margin Δ | $0.1\,\tau_ref$ |
| Decoded-Eikonal gap mask | $g > 3\tau$ |
| Sheet-consistency step $h_s$ | $0.05\,\tau_ref$ |

| Setting | Default |
|---|---|
| Tie threshold η / cosine threshold γ | 0.02 τ_ref / 0.95 |
| Near-cut batch fraction | 20 % |
| Usage check / dead threshold / patience | every 500 steps / 1 % / 3 checks |
| Revival noise / offset | 0.1 relative / 0.2 τ_ref |
| Merge condition | duplicate on more than 50 % of shared-band pairs |
| Commitment margin in planning | 0.05 τ_ref |
| Optimiser and encoder | the paper's best recipe (AdamW, cosine decay, Fourier σ as tuned per stage) |

**Do not:**

- train only the active branch; the cut is placed by the losing branch;
- define masks from predicted latents or predicted branch values without detaching them;
- plan with the soft-min gradient at the end of training; use the hard min and branch commitment;
- label branches by homotopy class or H-signature in version 1;
- tune hyperparameters on the test sets; select checkpoints on validation seeds only, as in the paper;
- describe the method as novel before checking it against min-plus networks (Darbon, Dower and Meng, 2021) and recent multi-branch distance-field work.